

# Research Plan: NLL Resummation of the Sudakov Shoulder in the C-Parameter

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## Abstract

This document presents a comprehensive research plan for deriving a factorization formula and performing next-to-leading logarithmic (NLL) resummation of the Sudakov shoulder in the C-parameter distribution for  $e^+e^-$  annihilation. The methodology follows and extends the analysis established for thrust and heavy jet mass by Bhattacharya, Schwartz, and Zhang. The C-parameter exhibits a *step discontinuity* at its LO phase space boundary  $C = 3/4$ , making it a particularly interesting case with unique features.

## Contents

# 1 Project Overview

## Project Goal

Derive and validate a factorization theorem and NLL resummed expression for the Sudakov shoulder in the C-parameter distribution in  $e^+e^-$  annihilation, following and extending the thrust/HJM shoulder analysis.

## 1.1 Core Ingredients Required

- (i) Precise characterization of the shoulder region at fixed order
- (ii) A SCET-based factorization theorem around the symmetric three-jet configuration
- (iii) One-loop hard/jet/soft functions and their anomalous dimensions
- (iv) NLL resummation, matching, and numerical implementation
- (v) Validation against fixed-order codes and analysis of systematic effects

## 1.2 Project Milestones

**M1:** Reproduce and understand the thrust/HJM shoulder analysis in the reference paper

**M2:** Complete NLO shoulder analysis for  $C$  and identify the logarithmic structure

**M3:** Derive and verify the factorization theorem for  $C$  in the shoulder limit

**M4:** Compute/assemble all one-loop ingredients and perform NLL resummation

**M5:** Implement numerical code and perform phenomenological/consistency checks

# 2 Observable Definition and Shoulder Kinematics

## C-Parameter Definition

The C-parameter is defined by:

$$C = \frac{3}{2} \sum_{i,j} \frac{|\vec{p}_i||\vec{p}_j|}{Q^2} \sin^2 \theta_{ij} = 3(\lambda_1 \lambda_2 + \lambda_2 \lambda_3 + \lambda_3 \lambda_1) \quad (1)$$

where  $\lambda_i$  are eigenvalues of the linearized momentum tensor  $\Theta^{\alpha\beta} = \frac{1}{\sum_k |\vec{p}_k|} \sum_j \frac{p_j^\alpha p_j^\beta}{|\vec{p}_j|}$  with  $\sum_i \lambda_i = 1$ .

At leading order for three massless partons:  $0 \leq C \leq C_{\text{sh}}$  with  $C_{\text{sh}} = 3/4$ .

### Key Insight: Discontinuity vs. Kink

Unlike thrust and heavy jet mass, which exhibit a **kink** (discontinuity in the first derivative) at their LO phase space boundary, the C-parameter has a **step discontinuity** in the distribution itself at  $C = 3/4$ . This fundamental difference may alter the structure of the resummed kernel.

## 2.1 Shoulder Variables

Define the shoulder variable for the right shoulder (NLO and beyond):

$$\Delta \equiv C - C_{\text{sh}} = C - \frac{3}{4} > 0 \quad (\text{right shoulder region}) \quad (2)$$

and for the left shoulder (where LO contributes):

$$c \equiv C_{\text{sh}} - C = \frac{3}{4} - C > 0 \quad (\text{left shoulder region}) \quad (3)$$

## 3 Stage I: Warm-Up and Kinematic Analysis

*Objective: Fix conventions, fully understand the geometry of the C-parameter shoulder, and benchmark against the thrust/HJM case.*

### 3.1 Literature and Reference Analysis

**Task 1.1:** Review the thrust/HJM shoulder paper (arXiv:2205.05702) in detail

**Task 1.2:** Review original C-parameter literature (Catani–Webber, hep-ph/9710333)

**Task 1.3:** Map notation and conventions between C-parameter literature and the reference paper

**Task 1.4:** Summarize similarities/differences: sensitivity to global soft radiation, recoil issues, number of hard directions

**Task 1.5:** Identify which arguments carry over trivially vs. which are C-specific

### 3.2 Three-Parton Kinematics

**Task 1.6:** Parameterize 3-body phase space using  $s_{ij} = (p_i + p_j)^2/Q^2$  with  $s_{12} + s_{13} + s_{23} = 1$

**Task 1.7:** Derive C in terms of invariants: verify  $C = 6 s_{12}s_{13}s_{23}$

**Task 1.8:** Confirm  $C_{\text{sh}} = 3/4$  at symmetric trijet:  $s_{12} = s_{13} = s_{23} = 1/3$

**Task 1.9:** Derive the LO distribution analytically and confirm step discontinuity behavior:

$$\frac{1}{\sigma_0} \frac{d\sigma}{dC} \propto \theta\left(\frac{3}{4} - C\right) \times f(C) \quad (4)$$

**Task 1.10:** Compute the explicit coefficient  $f(C)$  to set normalization for resummed calculation

### 3.3 Four-Parton Onset

**Task 1.11:** Characterize the first 4-parton configurations populating  $\Delta > 0$

**Task 1.12:** Introduce expansion parameter  $\lambda$  controlling deviations from symmetric trijet

**Task 1.13:** Relate  $\Delta$  (or  $c$ ) to soft/collinear kinematic variables at leading power

**Task 1.14:** Determine whether left shoulder, right shoulder, or both require resummation

## 4 Stage II: NLO Fixed-Order Analysis

*Objective: Isolate the regions of four-parton phase space that generate shoulder logarithms and extract their structure at  $\mathcal{O}(\alpha_s^2)$ .*

### 4.1 Phase-Space Classification

**Task 2.1:** Enumerate all 4-parton topologies:  $q\bar{q}gg$ ,  $q\bar{q}q'\bar{q}'$ ,  $q\bar{q}q\bar{q}$

**Task 2.2:** Separate regions continuously connected across  $C = 3/4$  from those that are not

**Task 2.3:** Identify which regions can produce large  $\ln \Delta$  through phase-space squeezing

**Task 2.4:** Derive phase space constraints analogous to  $\mathcal{W}(m_j, k_i, c) > 0$  from reference paper

### 4.2 Soft/Collinear Power Counting

**Task 2.5:** Define power counting: soft ( $p_s \sim Q\lambda$ ), collinear ( $p_c \sim Q(\lambda^2, 1, \lambda)$ ), soft-collinear

**Task 2.6:** Express  $C$  in terms of  $s_{ij}/Q^2$  and expand around symmetric trijet point

**Task 2.7:** Derive leading-power relation between  $\Delta$  and soft/collinear variables

**Task 2.8:** Document which integration regions contribute logarithms

### 4.3 Matrix Elements in Limiting Regions

**Task 2.9:** Compute  $|\mathcal{M}_{\gamma^* \rightarrow q\bar{q}gg}|^2$  in collinear limit with polarization structure

**Task 2.10:** Compute  $|\mathcal{M}_{\gamma^* \rightarrow q\bar{q}gg}|^2$  in soft limit with correct color structure

**Task 2.11:** Compute soft-collinear overlap for subtraction

**Task 2.12:** Handle  $q\bar{q}q'\bar{q}'$  and identical-quark channels

**Task 2.13:** Verify consistency with splitting functions

## 4.4 Extraction of NLO Shoulder Logarithms

**Task 2.14:** Perform leading-power phase-space integrals

**Task 2.15:** Extract  $\alpha_s^2 \ln^2 \Delta$  and  $\alpha_s^2 \ln \Delta$  coefficients

**Task 2.16:** Separate by color structure:  $C_F^2$ ,  $C_F C_A$ ,  $C_F T_F n_f$

**Task 2.17:** Run EVENT2 with high statistics ( $> 10^{12}$  events, cutoff  $\sim 10^{-12}$ )

**Task 2.18:** Compare analytic predictions to EVENT2 numerical results

**Task 2.19:** Create comparison plots analogous to Fig. 6 in reference paper

## 5 Stage III: Factorization Theorem in SCET

*Objective: Derive a SCET factorization theorem that captures the shoulder dynamics in terms of hard, jet, and soft functions.*

### 5.1 Mode Structure

**Task 3.1:** Specify three collinear directions  $n_i$  from LO 3-parton configuration

**Task 3.2:** Define scaling of collinear momenta in shoulder regime

**Task 3.3:** Define scaling of soft momenta

**Task 3.4:** Clarify whether collinear-soft or Glauber-like modes are required or can be excluded

**Task 3.5:** Contrast with dijet limit ( $C \rightarrow 0$ ) to ensure shoulder-specific kinematics captured

### 5.2 SCET Operator Basis

**Task 3.6:** Write leading-power SCET current for three hard partons

**Task 3.7:** Identify associated Wilson coefficient (hard function)

**Task 3.8:** Discuss mapping between 3-jet hard function for thrust/HJM and that needed for C

### 5.3 Measurement Function and Additive Decomposition

#### Key Insight: Additive Decomposition

A key step is deriving how the observable decomposes into collinear and soft contributions:

$$\Delta = f_{\text{coll}}(m_i^2/Q^2) + f_{\text{soft}}(k_s/Q) \quad (5)$$

This additive structure is essential for the convolution form of the factorization theorem.

**Task 3.9:** Express C in terms of SCET collinear and soft momenta at leading power

**Task 3.10:** Derive the additive decomposition relating  $\Delta$  to jet masses and soft radiation

**Task 3.11:** Define measurement functions for collinear and soft sectors separately

## 5.4 Factorized Cross Section

### Factorization Formula

$$\frac{d\sigma}{d\Delta} = H_C(Q, \mu) \int [dm_i^2][dk_s] \prod_{i=1}^3 J_i(m_i^2, \mu) S_C(k_s, \mu) \delta(\Delta - \mathcal{F}(m_i^2, k_s)) \quad (6)$$

where  $\mathcal{F}$  encodes the C-parameter measurement function.

**Task 3.12:** Derive the factorization formula with explicit measurement function

**Task 3.13:** Show hard 3-parton phase space factors from soft/collinear integrals

**Task 3.14:** Verify observable remains global at leading power (absence of non-global logs)

**Task 3.15:** Define 6-sextant decomposition of soft radiation (as in Fig. 4 of reference)

**Task 3.16:** Write 2-parameter trijet hemisphere soft function

## 5.5 Recoil Sensitivity Analysis

### Recoil Sensitivity Warning

Naive convolution pictures like  $\sigma(C) \sim \int dm^2 \sigma_{\text{LO}}(C - m^2) J(m^2)$  can be ambiguous beyond leading-log level. We must factorize *phase space* rather than just convolve with emission cross sections.

**Task 3.17:** Understand how soft/collinear emissions shift C-parameter value

**Task 3.18:** Analyze whether holding jet energy vs. momentum fixed gives different results

**Task 3.19:** Verify the measurement constraint  $\mathcal{F}$  properly captures kinematics without ambiguity

# 6 Stage IV: One-Loop Ingredients and Anomalous Dimensions

*Objective: Assemble all perturbative building blocks needed for NLL resummation.*

## 6.1 Hard Function

**Task 4.1:** Identify appropriate 3-jet hard function  $H_C(Q, \mu)$  and relation to thrust/HJM

**Task 4.2:** Collect one-loop expression and anomalous dimension from literature

**Task 4.3:** Translate conventions as needed

**Task 4.4:** Verify RG consistency (cusp and non-cusp pieces)

## 6.2 Jet Functions

**Task 4.5:** Determine if standard inclusive quark/gluon jet functions suffice

**Task 4.6:** If yes, import known one-loop expressions and anomalous dimensions

**Task 4.7:** If modifications needed, compute one-loop jet function with C-measurement

**Task 4.8:** Extract anomalous dimensions  $\gamma_J^q, \gamma_J^g$

## 6.3 Soft Function for C-Parameter Shoulder

**Task 4.9:** Define soft function  $S_C$  as matrix element of soft Wilson lines in 3 collinear directions

**Task 4.10:** Define measurement function  $M(k, q_i)$  with 6 sextant regions

**Task 4.11:** Specify projection vectors for each region

**Task 4.12:** Compute 4 independent integrals  $I_1, I_2, I_3, I_4$  (as in reference paper Fig. 5)

**Task 4.13:** Extract UV poles and finite parts using dimensional regularization

**Task 4.14:** Check for transcendental constants (Gieseking's constant  $\kappa$ , etc.)

**Task 4.15:** Combine with color structures for gluon-jet and quark-jet channels

**Task 4.16:** Extract cusp and non-cusp anomalous dimensions

## 6.4 RG Consistency Check

**Task 4.17:** Verify anomalous dimension sum rule:

$$\gamma_H + \gamma_{J_1} + \gamma_{J_2} + \gamma_{J_3} + \gamma_S = 0 \tag{7}$$

**Task 4.18:** Check whether soft function differs from thrust/HJM at NLL level

# 7 Stage V: NLL Resummation and Analytic Structure

*Objective: Perform NLL resummation of shoulder logarithms and analyze the resulting kernel.*

## 7.1 RG Evolution Kernels

**Task 5.1:** Write RG-evolved expressions for  $H_C, J_i, S_C$  with Sudakov exponents

**Task 5.2:** Retain cusp terms to two loops, non-cusp to one loop (NLL accuracy)

**Task 5.3:** Define Sudakov kernel  $S(\nu, \mu)$  and  $A_\Gamma(\nu, \mu)$

## 7.2 Canonical Scale Choices and Profile Functions

### Canonical Scales

$$\mu_h = Q, \quad \mu_j = \sqrt{\Delta} Q, \quad \mu_s = \Delta Q \quad (8)$$

These minimize logarithms in matching conditions and capture relevant physics at each scale.

**Task 5.4:** Verify canonical scales by examining log structure in each function

**Task 5.5:** Design profile functions for smooth transition to fixed-order region

**Task 5.6:** Implement scale variations for uncertainty estimation

## 7.3 Shoulder Kernel and Core Integrals

**Task 5.7:** Perform soft/collinear integrals with measurement function

**Task 5.8:** Obtain master kernel of form:

$$\Delta^{1+\eta_{\text{eff}}} \mathcal{K}_C(\eta_{\text{eff}}), \quad \eta_{\text{eff}} \propto \alpha_s \Gamma_0 \ln \Delta \quad (9)$$

**Task 5.9:** Compare to thrust integral (Eq. 129) and HJM integrals (Eqs. 130–131)

**Task 5.10:** Identify  $\sin(\pi\eta)/\sin(\pi(\eta_\ell + \eta_h))$  structure if present

**Task 5.11:** Handle UV/IR divergences via analytic continuation

## 7.4 Sudakov–Landau Pole Analysis

**Task 5.12:** Identify location of singularity  $\eta_\ell + \eta_h = 1$  in terms of  $\Delta$

**Task 5.13:** Estimate numerical value:  $\Delta_{\text{pole}} = \exp(-\text{const}/\alpha_s)$

**Task 5.14:** Compare to QCD Landau pole (different origin: cusp anomalous dimension vs.  $\beta$ -function)

**Task 5.15:** Determine valid range of  $\Delta$  for NLL predictions

## 7.5 Non-Global Logarithm Analysis

**Task 5.16:** Repeat shoulder-region analysis with auxiliary variables

**Task 5.17:** Show large logs arise only from regions continuously connected across shoulder

**Task 5.18:** Demonstrate configurations with unconstrained hard radiation don't generate  $\ln \Delta$

**Task 5.19:** Conclude whether non-global logs are absent at leading power

**Task 5.20:** Document any subleading-power non-global structures



## 7.6 Final NLL Formula

**Task 5.21:** Combine all ingredients into complete resummed expression

**Task 5.22:** Write separate formulas for gluon and quark channels

**Task 5.23:** Define subtracted distribution:

$$\frac{d\sigma^{\text{sub}}}{d\Delta} = \frac{d\sigma}{d\Delta} - \Delta \left( \frac{d\sigma}{d\Delta} \right)_{\Delta=1} \quad (10)$$

**Task 5.24:** Verify  $\mu$ -independence at  $\mathcal{O}(\alpha_s)$

## 8 Stage VI: Matching, Power Corrections, and Phenomenology

*Objective: Produce a usable prediction by matching to fixed order, estimating uncertainties, and comparing to data if available.*

### 8.1 Matching to Fixed Order

**Task 6.1:** Implement matching formula:

$$\left( \frac{d\sigma}{d\Delta} \right)_{\text{matched}} = \left( \frac{d\sigma}{d\Delta} \right)_{\text{res}} + \left( \frac{d\sigma}{d\Delta} \right)_{\text{FO}} - \left[ \left( \frac{d\sigma}{d\Delta} \right)_{\text{res}} \right]_{\text{expanded}} \quad (11)$$

**Task 6.2:** Decide NLO vs. NNLO fixed-order input

**Task 6.3:** Test matching in overlap region

**Task 6.4:** Verify smooth transition away from shoulder

### 8.2 Power Corrections

**Task 6.5:** Analyze subleading-power terms in soft/collinear expansion

**Task 6.6:** Show that for  $\eta \sim \mathcal{O}(1)$ , power-suppressed terms become  $\mathcal{O}(1)$

**Task 6.7:** Derive first subleading power contribution (analogous to Eq. 169 in reference)

**Task 6.8:** Demonstrate explicit pole cancellation when power corrections included

**Task 6.9:** Estimate window in  $\Delta$  where leading-power NLL is reliable

**Task 6.10:** Discuss  $\Lambda_{\text{QCD}}/Q$  non-perturbative corrections

### 8.3 Numerical Implementation

- Task 6.11:** Design modular code structure (Python or C++)
- Task 6.12:** Implement RG evolution kernels (reuse thrust/HJM code where possible)
- Task 6.13:** Implement hard, jet, and soft functions
- Task 6.14:** Implement shoulder convolution integrals
- Task 6.15:** Implement profile functions for scale transitions
- Task 6.16:** Include scale-variation machinery for uncertainties
- Task 6.17:** Validate against analytic limits

### 8.4 Validation and Cross-Checks

- Task 6.18:** Expand NLL resummed result to  $\mathcal{O}(\alpha_s^2)$  and  $\mathcal{O}(\alpha_s^3)$
- Task 6.19:** Compare to EVENT2/EERAD3 in shoulder limit
- Task 6.20:** Verify RG-scale independence at expected order
- Task 6.21:** Check all color structures separately

### 8.5 Comparison to Data (Optional)

- Task 6.22:** Generate predictions for LEP energies ( $\sqrt{s} = 91.2$  GeV, etc.)
- Task 6.23:** If suitable data exist near  $C = 3/4$ , perform exploratory comparisons
- Task 6.24:** Study theoretical uncertainties: scale variation, higher orders, power corrections
- Task 6.25:** Explore implications for  $\alpha_s$  extraction

## 9 Stage VII: Documentation and Extensions

*Objective: Document results and identify future directions.*

- Task 7.1:** Write up derivation of factorization theorem
- Task 7.2:** Document all one-loop results and anomalous dimensions
- Task 7.3:** Prepare figures comparing to fixed-order codes
- Task 7.4:** Discuss phenomenological implications
- Task 7.5:** Outline extensions: NNLL, joint resummation, other observables

| Stage              | Duration | Weeks | Key Deliverable                               |
|--------------------|----------|-------|-----------------------------------------------|
| I: Kinematics      | 2 weeks  | 1–2   | C-parameter in 4-parton kinematics            |
| II: NLO Analysis   | 3 weeks  | 3–5   | Extracted log coefficients, EVENT2 comparison |
| III: Factorization | 3 weeks  | 6–8   | Complete factorization formula                |
| IV: One-Loop       | 2 weeks  | 9–10  | 1-loop functions, anomalous dimensions        |
| V: Resummation     | 3 weeks  | 11–13 | NLL resummed formula, kernel analysis         |
| VI: Matching       | 3 weeks  | 14–16 | Matched predictions, numerical code           |
| VII: Documentation | 2 weeks  | 17–18 | Final writeup                                 |

Table 1: Project timeline with milestones (18 weeks total)

## 10 Timeline and Deliverables

## 11 Key Challenges and Open Questions

1. **Step vs. Kink:** The C-parameter has a step discontinuity at LO, unlike the kink for thrust/HJM. How does this affect the structure of shoulder logarithms and resummation?
2. **Measurement Function Complexity:** The C-parameter measurement may lead to more complex convolution integrals. What is the analog of the  $\sin(\pi\eta)$  structure?
3. **Additive Decomposition:** Can we cleanly separate collinear and soft contributions to  $\Delta$  as required for the factorization formula?
4. **Phenomenological Window:** The Sudakov–Landau pole limits the valid range. Is there a useful window for precision predictions?
5. **Connection to Catani–Webber:** How does the NLL formula connect to the original double-log resummation?
6. **Left vs. Right Shoulder:** Does C-parameter have structure on both sides of  $C = 3/4$ ? Which is phenomenologically relevant?
7. **Profile Functions:** What is the optimal profile function design for transitioning between resummation and fixed-order regions?
8.  $\alpha_s$  **Extraction:** Can shoulder resummation improve  $\alpha_s$  fits from C-parameter data?

## 12 Complete Task List

For reference, here is the complete ordered list of all tasks:

### Stage I: Warm-Up and Kinematic Analysis (Tasks 1.1–1.14)

- 1.1 Review the thrust/HJM shoulder paper (arXiv:2205.05702) in [detail](#)
- 1.2 Review original C-parameter literature (Catani–Webber, hep-ph/9710333)

- 1.3 Map notation and conventions between C-parameter literature and the reference paper
- 1.4 Summarize similarities/differences: sensitivity to global soft radiation, recoil issues
- 1.5 Identify which arguments carry over trivially vs. which are C-specific
- 1.6 Parameterize 3-body phase space using  $s_{ij}$  with constraint  $s_{12} + s_{13} + s_{23} = 1$
- 1.7 Derive C in terms of invariants: verify  $C = 6 s_{12} s_{13} s_{23}$
- 1.8 Confirm  $C_{\text{sh}} = 3/4$  at symmetric trijet
- 1.9 Derive the LO distribution analytically and confirm step discontinuity behavior
- 1.10 Compute the explicit coefficient  $f(C)$  to set normalization
- 1.11 Characterize the first 4-parton configurations populating  $\Delta > 0$
- 1.12 Introduce expansion parameter  $\lambda$  controlling deviations from symmetric trijet
- 1.13 Relate  $\Delta$  to soft/collinear kinematic variables at leading power
- 1.14 Determine whether left shoulder, right shoulder, or both require resummation

## Stage II: NLO Fixed-Order Analysis (Tasks 2.1–2.17)

- 2.1 Enumerate all 4-parton topologies
- 2.2 Separate regions continuously connected across  $C = 3/4$  from those that are not
- 2.3 Identify which regions can produce large  $\ln \Delta$
- 2.4 Derive phase space constraints analogous to  $\mathcal{W}(m_j, k_i, c) > 0$
- 2.5 Define power counting for soft, collinear, and soft-collinear modes
- 2.6 Express C in terms of  $s_{ij}/Q^2$  and expand around symmetric trijet point
- 2.7 Derive leading-power relation between  $\Delta$  and soft/collinear variables
- 2.8 Document which integration regions contribute logarithms
- 2.9 Compute  $|\mathcal{M}|^2$  in collinear limit with polarization structure
- 2.10 Compute  $|\mathcal{M}|^2$  in soft limit with correct color structure
- 2.11 Compute soft-collinear overlap for subtraction
- 2.12 Handle  $q\bar{q}q'\bar{q}'$  and identical-quark channels
- 2.13 Verify consistency with splitting functions
- 2.14 Perform leading-power phase-space integrals
- 2.15 Extract  $\alpha_s^2 \ln^2 \Delta$  and  $\alpha_s^2 \ln \Delta$  coefficients by color structure
- 2.16 Run EVENT2 with high statistics near  $C = 3/4$
- 2.17 Compare analytic predictions to EVENT2 and create comparison plots

### Stage III: Factorization Theorem (Tasks 3.1–3.16)

- 3.1 Specify three collinear directions  $n_i$  from LO configuration
- 3.2 Define scaling of collinear momenta in shoulder regime
- 3.3 Define scaling of soft momenta
- 3.4 Clarify whether collinear-soft or Glauber modes are required
- 3.5 Contrast with dijet limit to ensure shoulder-specific kinematics captured
- 3.6 Write leading-power SCET current for three hard parts
- 3.7 Identify associated Wilson coefficient (hard function)
- 3.8 Discuss mapping to thrust/HJM hard function
- 3.9 Express C in terms of SCET momenta at leading power
- 3.10 Derive the additive decomposition for  $\Delta$
- 3.11 Define measurement functions for collinear and soft sectors
- 3.12 Derive the factorization formula with explicit measurement function
- 3.13 Verify observable remains global at leading power
- 3.14 Define 6-sextant decomposition of soft radiation
- 3.15 Analyze recoil sensitivity: how emissions shift C-parameter value
- 3.16 Verify measurement constraint captures kinematics without ambiguity

### Stage IV: One-Loop Ingredients (Tasks 4.1–4.14)

- 4.1 Identify 3-jet hard function and relation to thrust/HJM
- 4.2 Collect one-loop hard function expression and anomalous dimension
- 4.3 Translate conventions as needed
- 4.4 Verify hard function RG consistency
- 4.5 Determine if standard jet functions suffice
- 4.6 Import or compute one-loop jet functions
- 4.7 Extract jet function anomalous dimensions
- 4.8 Define soft function with 6 sextant measurement
- 4.9 Specify projection vectors for each sextant region
- 4.10 Compute 4 independent soft integrals  $I_1, I_2, I_3, I_4$

- 4.11 Extract UV poles, finite parts, and transcendental constants
- 4.12 Combine with color structures for all channels
- 4.13 Verify anomalous dimension sum rule:  $\gamma_H + \sum \gamma_J + \gamma_S = 0$
- 4.14 Check whether soft function differs from thrust/HJM at NLL

## Stage V: NLL Resummation (Tasks 5.1–5.18)

- 5.1 Write RG-evolved expressions with Sudakov exponents
- 5.2 Define Sudakov kernel to NLL accuracy
- 5.3 Verify canonical scales by examining log structure
- 5.4 Design profile functions for transition to fixed-order region
- 5.5 Implement scale variations for uncertainties
- 5.6 Perform soft/collinear integrals with measurement function
- 5.7 Obtain master kernel  $\Delta^{1+\eta_{\text{eff}}} \mathcal{K}_C(\eta_{\text{eff}})$
- 5.8 Compare to thrust/HJM kernel structure
- 5.9 Identify  $\sin(\pi\eta)$  structure if present
- 5.10 Handle UV/IR divergences via analytic continuation
- 5.11 Identify Sudakov–Landau pole location
- 5.12 Estimate numerical value of  $\Delta_{\text{pole}}$
- 5.13 Determine valid range of  $\Delta$  for NLL predictions
- 5.14 Analyze non-global logarithm structure
- 5.15 Conclude whether non-global logs are absent at leading power
- 5.16 Combine all ingredients into complete resummed expression
- 5.17 Write separate formulas for gluon and quark channels
- 5.18 Verify  $\mu$ -independence at  $\mathcal{O}(\alpha_s)$

## Stage VI: Matching and Phenomenology (Tasks 6.1–6.18)

- 6.1 Implement matching formula
- 6.2 Decide NLO vs. NNLO fixed-order input
- 6.3 Test matching in overlap region
- 6.4 Verify smooth transition away from shoulder

- 6.5 Analyze subleading-power terms
- 6.6 Show power-suppressed terms become  $\mathcal{O}(1)$  when  $\eta \sim \mathcal{O}(1)$
- 6.7 Derive first subleading power contribution
- 6.8 Demonstrate explicit pole cancellation with power corrections
- 6.9 Estimate window where leading-power NLL is reliable
- 6.10 Discuss non-perturbative corrections
- 6.11 Design modular numerical code
- 6.12 Implement RG evolution kernels
- 6.13 Implement hard, jet, and soft functions
- 6.14 Implement shoulder convolution integrals
- 6.15 Validate code against analytic limits
- 6.16 Expand NLL result and compare to EVENT2/EERAD3
- 6.17 Verify RG-scale independence
- 6.18 Generate predictions and compare to data if available

## Stage VII: Documentation (Tasks 7.1–7.5)

- 7.1 Write up derivation of factorization theorem
- 7.2 Document all one-loop results and anomalous dimensions
- 7.3 Prepare figures comparing to fixed-order codes
- 7.4 Discuss phenomenological implications
- 7.5 Outline extensions: NNLL, joint resummation, other observables